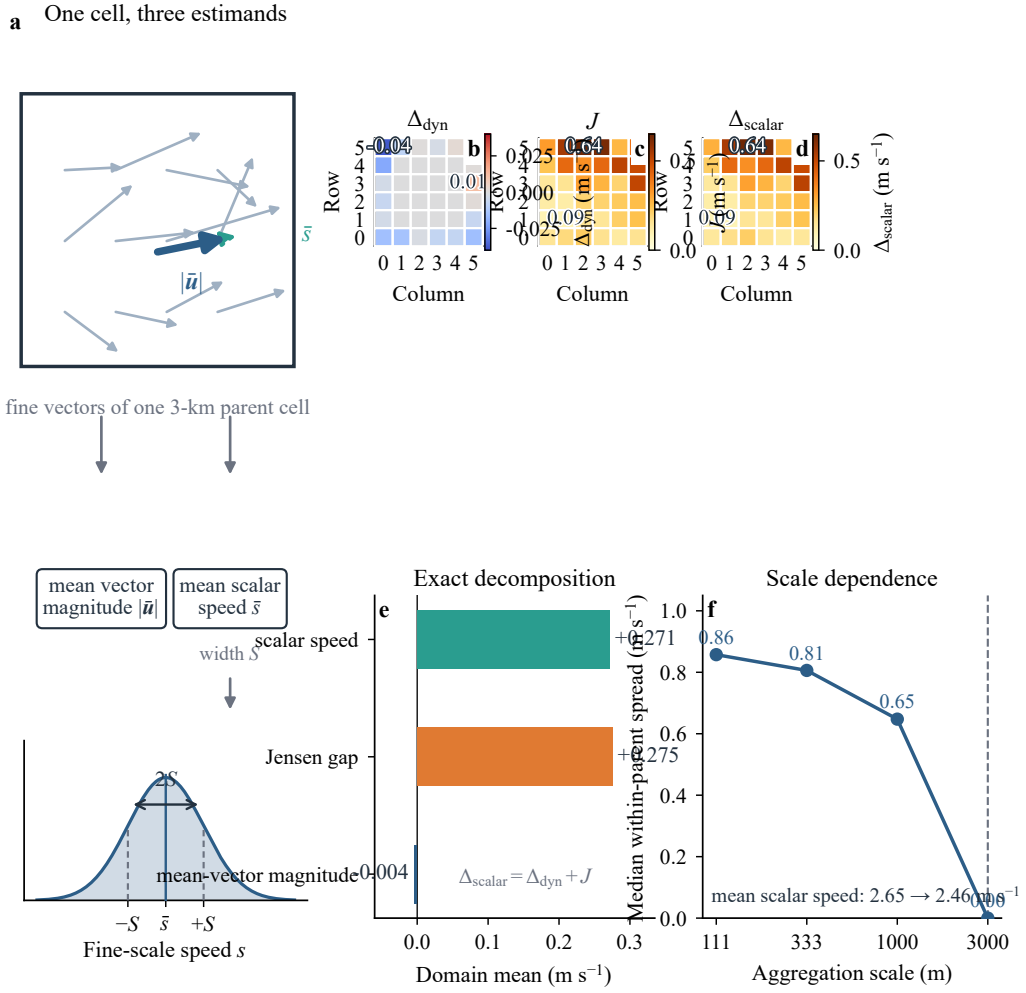

a
One cell, three estimands
$|\bar{u}|$
fine vectors of one 3-km parent cell
Δ_{dyn}
J
Δ_{scalar}
b
c
d
Row
Column
Δ_{dyn} (m s^{-1})
J (m s^{-1})
Δ_{scalar} (m s^{-1})
mean vector magnitude $|\bar{u}|$
mean scalar speed $\bar{s}$
width S
$2S$
mean-vector magnitude
$-S$
$\bar{s}$
$+S$
Fine-scale speed s
Exact decomposition
e
scalar speed
Jensen gap
+0.271
+0.275
-0.004
$\Delta_{scalar} = \Delta_{dyn} + J$
Domain mean (m s^{-1})
Scale dependence
f
Median within-parent spread (m s^{-1})
0.86
0.81
0.65
0.00
mean scalar speed: 2.65 → 2.46 m s^{-1}
Aggregation scale (m)
111
333
1000
3000